

192.151 Introduction to Deep Learning

2026S

Project 5: Image Captioning

1. Background and Motivation

Image captioning bridges computer vision and natural language processing, aiming to create systems that can “describe” an image in natural language. This task requires understanding the visual content of an image (objects, attributes, actions, and their relationships) and then generating a grammatically correct and semantically relevant textual description. This project will explore a common encoder-decoder architecture, often using a CNN as an encoder and an RNN (like LSTM or GRU) as a decoder.

2. Problem Description

The main goal is to implement an image captioning model. Given an image, the model should output a plausible caption. This involves:

a) Dataset Preparation:

- Select an image captioning dataset (e.g., a subset of Flickr8k, COCO).
- Implement image preprocessing and caption tokenization.

b) Model Implementation:

- Implement the CNN encoder to extract image features (e.g., VGG or ResNet).
- Implement the RNN decoder for caption generation (e.g., LSTM or GRU), including word embeddings.
- Connect the encoder and decoder appropriately.

c) Training and Evaluation:

- Train the image captioning model.
- Generate captions for test images and qualitatively assess them.
- Evaluate using BLEU scores or other captioning metrics.
- Experiment with hyperparameters (e.g., embedding size, RNN hidden units, learning rate).
- (Optional) explore attention mechanisms to allow the decoder to focus on relevant image regions.

3. Expected Deliverables

Students working in groups (ideally 3 members) are expected to:

- produce a project report,
- deliver a group presentation, and
- submit well-commented source code.

4. Suggested Resources and References

- [1] Vinyals, O., Toshev, A., Bengio, S., & Erhan, D. (2015). Show and Tell: A Neural Image Caption Generator. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 3156–3164). <https://arxiv.org/abs/1411.4555>
- [2] Karpathy, A., & Fei-Fei, L. (2015). Deep Visual-Semantic Alignments for Generating Image Descriptions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 3128–3137). <https://arxiv.org/abs/1412.2306>
- [3] Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., & Bengio, Y. (2015). Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. In *Proceedings of the International Conference on Machine Learning (ICML)* (pp. 2048–2057). <https://arxiv.org/abs/1502.03044>